Benchmark results

unknown AArch64 TOS 4.04 640x480 256c blit none

115 of 115 tests run, 1.0 s each. Values are operations per second (higher is better). Bars compare tests within each group.

Groups

VDI primitives	38 tests run
AES windows & events	21 tests run
Alignment (blitter)	20 tests run
Full-screen windows	17 tests run
GDOS fonts	14 tests run

VDI primitives

Test	ops/s	Note
Line, 2 points	54873	
Polyline, 32 points	2910.9	
Wide line (7px, round)	12923	
Dashed line	48906	
User line style	49763	
XOR line	54612	
Markers x16	43114	
Filled rect 32x32	85820	
Filled rect, full window	692.6	
Bar, hatched + border	52202	
Pattern fill rect	85010	
User pattern fill	85820	
XOR filled rect	80708	
Polygon, 8 points	29256	
Filled circle	7348.3	
Filled ellipse	7348.3	
Arc	61825	
Pie slice	7210.3	
Elliptical arc	61825	
Elliptical pie	9845.1	
Rounded box	91871	
Filled rounded box	24452	
Contour fill 64x40	5312.2	
Text, 15 chars	31506	
Text, 84 chars	7691.0	
Text with effects	14123	
Text rotated 90	24575	
Justified text	18069	
Text extent query	276911	
Blit scr->scr 64x64	21660	
Blit scroll window	469.6	
Blit mem->scr 64x64	33360	
Blit scr->mem 64x64	29537	
Mono blit (vrt_cpyfm)	13334	
Transform form 64x64	n/a	not supported here
Get pixel	333747	
Hide/show mouse	38279	
Attribute calls x3	120469	
Open/close workstation	24881	

AES windows & events

Test	ops/s	Note
Window create+delete	120469	
Window open+close	347.5	
Window move	1200.9	
Window resize	1554.2	
Window top (2 windows)	703.0	
Set window title	4226.6	
Set info line	4265.7	
Set slider pos+size	8776.1	
Scroll window content	451.8	
wind_get work area	280867	
Rectangle list walk	122879	
wind_calc	223780	
wind_update pair	138455	
objc_draw dialog	5493.6	
objc_find	202848	
form_dial redraw 64x48	6143.0	
Menu bar redraw	6248.8	
graf_mkstate	282211	
graf_mouse off+on	35616	
Event poll (evnt_multi)	114687	
Message round trip	48906	

Alignment (blitter)

Test	ops/s	Note
Fill 64x64, aligned	28671	
Fill 64x64, x+3	29117	
Blit 64x64, aligned	20666	
Blit 64x64, src+3 dst+11	21053	
Mem->scr 64x64, aligned	33209	
Mem->scr 64x64, x+3	32442	
Mono blit, aligned	13783	
Mono blit, x+3	13652	
Text, x on 8 px	32124	
Text, x+3	31506	
Scroll up, aligned edges	529.9	
Scroll up, edges x+5	529.9	
Scroll left 16 px	539.9	
Scroll left 3 px	522.7	
Move window, aligned	1055.9	work x mod 16 = 0
Move window, off-grid	1041.6	work x mod 16 = 5
Open/close win, aligned	340.7	work x mod 16 = 0
Open/close win, off-grid	340.7	work x mod 16 = 5
Scroll window, aligned	1583.8	work x mod 16 = 0
Scroll window, off-grid	1568.8	work x mod 16 = 5

Full-screen windows

Test	ops/s	Note
Open/close full window	180.1	work 616x418
Full <-> normal size	342.4	
Top full / test window	412.9	work 616x418
Fill full window	444.7	work 616x418
Scroll up 1 line	296.7	work 616x418
Scroll left 16 px	296.7	work 616x418
Scroll left 8 px	298.1	work 616x418
Scroll left 3 px	298.1	work 616x418
Scroll text 8pt	300.9	system font
Scroll text 10pt	298.1	system font
Scroll text 12pt	295.3	got 10pt, system font
Scroll text 14pt	296.7	got 10pt, system font
Scroll text 18pt	287.0	system font
Scroll text 24pt	278.6	got 20pt, system font
Scroll text 36pt	280.0	got 20pt, system font
Scroll text 48pt	278.6	got 20pt, system font
Scroll, size per line	288.6	system font

GDOS fonts

Test	ops/s	Note
Text 6pt	50233	got 8pt, system font
Text 8pt	50233	system font
Text 10pt	22649	system font
Text 12pt	22860	got 10pt, system font
Text 14pt	22649	got 10pt, system font
Text 18pt	10380	system font
Text 24pt	5575.2	got 20pt, system font
Text 36pt	5687.9	got 20pt, system font
Text 48pt	5661.7	got 20pt, system font
Text 72pt	5687.9	got 20pt, system font
Text 10pt bold	21977	system font
vst_point, 8 sizes	264494	
Font switch + vst_point	n/a	not supported here
Arbitrary size + text	n/a	not supported here
Scroll, mixed sizes	431.8	system font
Text extent at 12pt	271807	
Enumerate font names	n/a	not supported here
Load+unload fonts (x3)	n/a	not supported here

Configuration

Machine

```
Description     : (not set - Options menu)
Machine type    : unknown
CPU             : AArch64 (native)
FPU             : none reported
Blitter         : not present
ST RAM          : 14336 KB total
TT/Fast RAM     : 925696 KB total
```

Operating system

```
TOS             : 4.04  (11/06/2026 m/d/y)
Country / video : USA, NTSC
GEMDOS          : 0.48
MiNT            : not running
Largest free    : ST 10240 KB
                  TT/Fast 914980 KB
```

AES / environment

```
AES version     : 3.40
Multitasking    : no (max apps 1)
Theme pens      : green 166, red 122, blue 149, cyan 155, highlight 106
Application id  : 0
appl_getinfo    : not available
Desktop area    : 0,19 640x461
AES char / box  : 8x16 / 19x19
Current dir     : C:\
PATH            : /dev/c/
```

Display

```
Resolution      : 640x480, Getrez 4 (TT Medium)
Colours         : 256 on screen, 16 planes
Palette         : 0 (direct colour)
Lookup table    : no (true colour)
Pixel size      : 278 x 278 microns
Screen memory   : 614400 bytes at $3c100000
VDI handles     : physical 1, ui 2 (bench ws opened only for runs)
VDI max vertices: 1024
Text rotation   : 90 degree steps
Writing modes   : 4
```

GDOS / fonts

```
GDOS            : none (system font only)
Fonts           : 1 (0 loaded by GDOS)
  > id     1  6x6 system font
```

Cookie jar

```
NTOS _MCH _IDT _AKP EdDI
```